

Domain-Generalizable Phishing URL Detection under Realistic Evaluation Splits

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Abstract

Phishing websites copy real websites to steal passwords and money. Most machine-learning tools that catch phishing links are tested the easy way: the data is shuffled first, so the same website family can end up in both training and testing. This makes the tool look better than it really is. We test detectors the honest way, using test websites from domains the model has never seen, and measure how much accuracy drops once the shortcuts are removed.

Keywords

phishing detection, domain generalization, evaluation methodology, machine learning

1 Introduction and Motivation

Phishing is when a fake link pretends to be a trusted website to steal someone’s login or payment details. In 2025 alone, security groups tracked over 3.8 million phishing attacks [2]. A good detector could catch these links before anyone clicks them. Past work already reports very accurate results [3], but that only matters if the detector also catches brand-new attacker websites, not just ones like its training data.

1.1 Motivating Example

Imagine a message that says: “Your PayPal account is locked. Fix it here: <https://paypal.login-fraud.top>.” The word “paypal” sits in the link, but the real website behind it is login-fraud.top, a domain that has never been seen before. A useful detector needs to catch links like this even though it never trained on that exact website.

2 Gap and Project Objective

Most testing methods split data randomly, so subdomains of one company, such as mail.paypal.com, can land in both sets. Then the model is not really tested on anything new because it already learned about that company. Cross-dataset transfer is difficult enough that a specialized domain-adaptation method recovered only 0.06 F1 on average and up to 0.2 in the best case [4]. Even a newer 2026 method that splits by host still allowed 1,387 matching domains to leak across [1].

We will use the registrable domain (for example, paypal.com) as the one thing that must never repeat between training and testing. We compare a random split, a host-disjoint split, a registrable-domain-disjoint split, and a cross-source split on a second dataset. Experiments use the 235,795-site PhiUSIIL corpus [3] plus an independent public URL corpus. Our contribution is a fair, reusable test protocol and a real number for how much the old style of testing overstated performance.

3 Formal Problem Definition

In plain words, given a link, decide whether it is phishing or safe without ever training on that link’s own website. Formally, let $\mathcal{U}$ be the set of all URLs, $\mathcal{Y} = \{0, 1\}$, where 1 means phishing and 0 means safe, and let $r : \mathcal{U} \rightarrow \mathcal{R}$ map a URL to its registrable domain. Given a labeled training set $D_{\text{tr}} = \{(u_i, y_i)\}_{i=1}^n \subset \mathcal{U} \times \mathcal{Y}$, the task is to learn a function

$$f : \mathcal{U} \rightarrow \mathcal{Y}$$

that correctly labels a new URL u whose registrable domain never appeared in training:

$$r(u) \notin \{r(u_i) : (u_i, y_i) \in D_{\text{tr}}\}.$$

In the cross-source setting, test URLs also come from a collection process never used for D_{tr} .

3.1 Example Instance

For $u = \text{https://paypal.login-fraud.top}$, suppose login-fraud.top is absent from training. The expected output is $f(u) = 1$ (phishing).

4 Team and Collaboration

Prabhnoor checks the data sources and keeps the test splits leakage-free. Divraj runs the experiments and keeps results reproducible. Dilraj combines the datasets, tracks experiments, manages the code repository, and studies where the model errs. All three work together on the problem definition, related work, code, experiments, and writing, with complementary data-engineering, machine-learning, and cybersecurity backgrounds. Repository: <https://github.com/prabhnoob/domain-generalizable-phishing-url-detection>.

References

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